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SEMICONDUCTOR DEVICES AND MANUFACTURING METHODS THEREOF

Abstract

Examples of the present disclosure provide a semiconductor device and a manufacturing method thereof. The semiconductor device includes semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; gate lines spaced apart along the first direction, wherein each of the gate lines extends along the second direction and is connected with the semiconductor pillars arranged along the second direction; a first cavity between two adjacent gate lines along the first direction and extending along the second direction; bit lines spaced apart along the second direction, wherein each of the bit lines extends along the first direction and is connected with the semiconductor pillars arranged along the first direction; and a second cavity between two adjacent bit lines along the second direction and extending along the first direction, wherein the second cavity and the first cavity are connected.

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Background/Summary

REFERENCE TO RELATED APPLICATION

[0001] The present application claims the benefit of priority to China Application No. 202410257114.8, filed on Mar. 5, 2024, the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Examples of the present disclosure relate to the field of semiconductor technology, and particularly to semiconductor devices and manufacturing methods thereof.

BACKGROUND

[0003] A semiconductor device, e.g. a Dynamic Random Access Memory (DRAM), generally comprises a plurality of memory cells, each of which may comprise one transistor and one memory capacitor, wherein a gate of the transistor is connected with a word line, a source or a drain of the transistor is connected with the memory capacitor, and the drain or the source of the transistor is connected with a bit line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIGS. 1A to 1D are cross-sectional structural diagrams of a semiconductor device provided by an example;

[0005] FIG. 2 is a flow diagram of a manufacturing method of a semiconductor device provided by examples of the present disclosure;

[0006] FIGS. 3A to 3G are cross-sectional structural diagrams of a semiconductor device provided by a first example in a manufacturing process;

[0007] FIGS. 4A to 4G are cross-sectional structural diagrams of a semiconductor device provided by a second example in a manufacturing process;

[0008] FIGS. 5A to 5G are cross-sectional structural diagrams of a semiconductor device provided by a third example in a manufacturing process;

[0009] FIGS. 6A to 6G are cross-sectional structural diagrams of a semiconductor device provided by a fourth example in a manufacturing process;

[0010] FIGS. 7A to 7C are cross-sectional structural diagrams of a semiconductor device provided by a first example;

[0011] FIGS. 8A to 8C are cross-sectional structural diagrams of a semiconductor device provided by a second example;

[0012] FIGS. 9A to 9C are cross-sectional structural diagrams of a semiconductor device provided by a third example; and

[0013] FIGS. 10A to 10C are cross-sectional structural diagrams of a semiconductor device provided by a fourth example.

DETAILED DESCRIPTION

[0014] The technical solutions in implementations of the present disclosure will be described below clearly and completely in conjunction with the implementations and the drawings of the present disclosure. Apparently, the implementations described are only part, but not all, of the implementations of the present disclosure. All other implementations obtained by those of ordinary

skills in the art based on the implementations in the present disclosure without creative work shall fall in the scope of protection of the present disclosure.

[0015] In the description below, many specific details are presented to provide a more thorough understanding of the present disclosure. However, it is apparent to a person skilled in the art that the present disclosure may be implemented without one or more of these details. In other examples, in order to avoid confusion with the present disclosure, some technical features well-known in the field are not described. That is, all the features of the actual examples are not described here, and well-known functions and structures are not described in detail.

[0016] In the drawings, the sizes of a layer, a region, and an element and their relative sizes may be exaggerated for clarity. The same reference numeral represents the same element throughout.

[0017] It is to be understood that when an element or a layer is referred to as being “on”, “adjacent to”, “connected to”, or “coupled to” other elements or layers, it may be directly on, adjacent to, connected to, or coupled to the other elements or layers, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on”, “directly adjacent to”, “directly connected to”, or “directly coupled to” other elements or layers, no intervening elements or layers are present. It should be understood that, although the terms first, second, third, etc., may be used to describe various elements, components, areas, layers and/or parts, these elements, components, areas, layers and/or parts should not be limited by these terms. These terms are only used to distinguish one element, component, area, layer or part from another element, component, area, layer or part. Thus, a first element, component, area, layer or part discussed below may be represented as a second element, component, area, layer or part, without departing from the teachings of the present disclosure. However, when the second element, component, area, layer or part is discussed, it does not mean that the first element, component, area, layer or part is necessarily present in the present disclosure.

[0018] The spatially relative terms, such as “beneath”, “below”, “lower”, “under”, “above”, “upper”, and the like, may be used herein for ease of description to describe the relationship of one element or feature to other elements or features as illustrated in the figures. It is to be understood that, the spatially relative terms are intended to further include different orientations of a device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the drawings is flipped, the elements or the features described as “below” or “under” or “beneath” other elements may be oriented “on” the other elements or features. Thus, the example terms, “below” and “beneath”, may comprise both upper and lower orientations. The device may be orientated otherwise (rotated by 90 degrees or other orientations), and the spatially descriptive terms used herein are interpreted accordingly.

[0019] The terms used herein are only intended to describe the specific examples, and are not used as limitations of the present disclosure. As used herein, unless otherwise indicated expressly in the context, “a”, “an” and “the” in a singular form are also intended to comprise a plural form. It should also be understood that the terms “consist of” and/or “comprise”, when used in this specification, determine the presence of the feature, integer, step, operation, element and/or component, but do not preclude the presence or addition of one or more of other features, integers, steps, operations, elements, components, and/or groups. As used herein, the term “and/or” includes any and all combinations of related items listed.

[0020] In order to understand the present disclosure thoroughly, detailed steps and detailed structures will be proposed in the following description to set forth the technical solution of the present disclosure. The detailed descriptions of some examples of the present disclosure are as follows. However, the present disclosure may also have other implementations in addition to these detailed descriptions.

[0021] Currently, the manufacturing process of DRAM is relatively complicated, which may reduce its productivity and yield. Therefore, how to improve the manufacturing process of the semiconductor device to improve the performance of the semiconductor device has become an

urgent problem to be solved.

[0022] With the development of DRAM technology, the size of memory cell is increasingly smaller, and its array architecture varies from 8 F.sup.2 to 6 F.sup.2 and then to 4 F.sup.2. In addition, based on requirements of the DRAM on ions and leakage current, the architecture of the memory varies from a planar array transistor to a recess gate array transistor, then from the recess gate array transistor to a buried channel array transistor, and then from the buried channel array transistor to a vertical channel array transistor.

[0023] In some examples, for either the planar transistor or the buried transistor, the DRAM is composed of a plurality of memory cells each being composed of one transistor and one memory capacitor controlled by the transistor, that is, the DRAM comprises an architecture of one transistor and one memory capacitor (One Transistor One Capacitor, 1T1C). Its main operation principle is to represent whether a binary bit is 1 or 0 according to the amount of charges stored in the memory capacitor.

[0024] One of the architectures of the DRAM is described in detail below in conjunction with the drawings. It should be understood that the following description with respect to the DRAM architecture is only used to illustrate the present disclosure, and is not used to limit the scope of the present disclosure.

[0025] Before introducing the examples of the present disclosure, first, various directions that may be used to describe the semiconductor device in the examples of the present disclosure are defined. A thickness direction of a first substrate (or a second substrate) is defined as a Z direction, that is, a direction perpendicular to the first substrate is defined as the Z direction. An X direction and a Y direction that intersect with each other are defined in a plane perpendicular to the Z direction, that is, the X direction and the Y direction are both parallel to the first substrate (or the second substrate). In some examples, the X direction and the Y direction may be perpendicular to each other. In some other examples, the X direction and the Y direction may be also not perpendicular to each other. In the examples of the present disclosure, a first direction is the X direction, a second direction is the Y direction, a third direction is the Z direction, and an illustration will be made in the following examples by taking the X direction, the Y direction and the Z direction mutually perpendicular to each other as an example.

[0026] With reference to FIG. 1A, FIG. 1A is a cross-sectional structural diagram of a semiconductor device provided by an example. As shown in FIG. 1A, the semiconductor device **100** comprises a first semiconductor structure **102** and a second semiconductor structure **118** stacked over the first semiconductor structure **102** along a Z direction, wherein the first semiconductor structure **102** is connected with the second semiconductor structure **118** through a bonding interface **116**, and the first semiconductor structure **102** may be connected with the second semiconductor structure **118** by means of hybrid bonding, etc. In some examples, the second semiconductor structure **118** may be bonded over the first semiconductor structure **102** at the bonding interface **116** in a face-to-face manner.

[0027] The first semiconductor structure **102** may comprise a first substrate **104**, a peripheral circuit **106** on a side of the first substrate **104**, and a first interconnect layer **110** on a side of the peripheral circuit **106** away from the first substrate **104**, wherein the first interconnect layer **110** is used to transfer electrical signals of the peripheral circuit **106**. The peripheral circuit **106** may comprise a plurality of transistors **108**. In some examples, a trench isolation structure and a doped region (e.g., a well, a source and a drain of the transistor **108**) may be also formed over or in the first substrate **104**.

[0028] The first semiconductor structure **102** may further comprise a first bonding layer **112** at the bonding interface **116** and on a side of the first interconnect layer **110** away from the peripheral circuit **106**. The first bonding layer **112** may comprise a plurality of first bonding contacts **114** and a dielectric layer electrically isolating the first bonding contacts **114**. The first bonding contacts **114** and the surrounding dielectric layer in the first bonding layer **112** may be used for hybrid bonding.

In contrast, the second semiconductor structure **118** may also comprise a second bonding layer **120** at the bonding interface **116** and on a side of the first bonding layer **112** away from the first interconnect layer **110**. The second bonding layer **120** may comprise a plurality of second bonding contacts **122** and a dielectric layer electrically isolating the second bonding contacts **122**. The second bonding contacts **122** and the surrounding dielectric layer in the second bonding layer **120** may be used for hybrid bonding. Here, the second bonding contacts **122** are in contact with the first bonding contacts **114** at the bonding interface **116**.

[0029] In some examples, the peripheral circuit **106** may further comprise a word line (WL) (also referred to as a gate line) and a word line driver/row decoder coupled to a second interconnect layer **124** through the second bonding contacts **122** in the second bonding layer **120** and the first bonding contacts **114** in the first bonding layer **112** and the first interconnect layer **110**. In some other examples, the peripheral circuit **106** may further comprise a bit line (BL) and a bit line driver/column decoder coupled to the second interconnect layer **124** through the second bonding contacts **122** in the second bonding layer **120** and the first bonding contacts **114** in the first bonding layer **112** and the first interconnect layer **110**.

[0030] With reference to FIG. **1A**, the second semiconductor structure **118** further comprises a memory cell array over the second interconnect layer **124**, wherein the memory cell array may comprise a plurality of memory cells **128**, a second substrate **156** over the memory cells **128**, and a third interconnect layer **158** over the second substrate **156**. Here, the second interconnect layer **124** comprises a bit line **126** above the second bonding layer **120**, wherein the bit line **126** is used for transferring electrical signals. A cross section of the semiconductor device **100** of FIG. **1A** may be taken along a bit line direction (i.e., the X direction), and one bit line **126** in the second interconnect layer **124** extending laterally in the X direction may be coupled to a row of vertical transistors **130**.

[0031] Here, each memory cell **128** may comprise a vertical transistor **130** and a memory capacitor coupled to the vertical transistor **130**. The vertical transistor **130** comprises a semiconductor pillar **134** extending vertically (i.e., the Z direction), and a gate structure **140** in the bit line direction (i.e., the X direction) and in contact with a side of the semiconductor pillar **134**. In some other examples, the gate structure may also be all around the semiconductor pillar, half around the semiconductor pillar, on two opposite sides of the semiconductor pillar, and the like, which is no longer repeated here. Here, the gate structure **140** comprises a gate layer **138**, and a gate dielectric layer **136** between the gate layer **138** and the semiconductor pillar **134**.

[0032] In some examples, the semiconductor pillar **134** has two ends (an upper end and a lower end) in the vertical direction (i.e., the Z direction), and at least one end (e.g., the lower end in FIG. **1A**) extends beyond the gate dielectric layer **136** into an interlayer dielectric (ILD) layer in the vertical direction (i.e., the Z direction). In some examples, one end (e.g., the upper end in FIG. **1A**) of the semiconductor pillar **134** is flush with a respective end of the gate dielectric layer **136**. In some examples, the two ends (the upper end and the lower end) of the semiconductor pillar **134** extend beyond the gate layer **138** into the ILD in the vertical direction (i.e., the Z direction) respectively. That is, the semiconductor pillar **134** may have a vertical size greater than a vertical size (e.g., a height in the Z direction) of the gate layer **138**, and both the upper end and the lower end of the semiconductor pillar **134** are not flush with the respective ends of the gate layer **138**. As such, a short circuit between the bit line **126** and the gate layer **138** or between the gate layer **138** and the memory capacitor **132** may be avoided.

[0033] The vertical transistor **130** may further comprise a source **142** and a drain **144** (locations of the source and the drain may be interchangeable) disposed at the two ends (the upper end and the lower end) of the semiconductor pillar **134** in the Z direction respectively. In some implementations, the source **142** is coupled to the memory capacitor **132**, and the drain **144** is coupled to the bit line **126**.

[0034] Since the gate layer **138** may be part of a gate line or extend in a gate line direction as a gate

line, the second semiconductor structure **118** of the semiconductor device **100** may also comprise a plurality of gate lines each extending in the gate line direction (i.e., the Y direction). Here, each gate line may be coupled to a column of vertical transistors **130**.

[0035] The vertical transistor **130** vertically extends through and is in contact with the gate line, and the drain **144** of the vertical transistor **130** at the lower end thereof is in contact with the bit line **126**. Therefore, due to the vertical arrangement of the vertical transistor **130**, the gate line and the bit line **126** may be disposed in different planes in the vertical direction, which simplifies routing of the gate line and the bit line **126**. Here, the vertical transistor **130** may be arranged in a mirror-symmetrical manner so as to increase the density of the memory cells **128** in the bit line direction (i.e., the X direction). The two adjacent vertical transistors **130** in the bit line direction are mirror-symmetrical to each other with respect to a trench isolation structure **164**, that is to say, the second semiconductor structure **118** may comprise a plurality of trench isolation structures **164**, wherein each trench isolation structure **164** and the gate line extend in parallel in the gate line direction (i.e., the Y direction), and are disposed between two adjacent columns of semiconductor pillars **134** of the vertical transistor **130**. In some implementations, the columns of vertical transistors **130** separated by the trench isolation structure **164** are mirror-symmetrical to each other with respect to the trench isolation structure **164**. It should be understood that the trench isolation structure **164** may comprise air gaps each laterally disposed between the two adjacent semiconductor pillars **134**.

[0036] As shown in FIG. 1A, the memory capacitor **132** comprises a first electrode **148** above the source **142** (e.g., the upper end of the semiconductor pillar **134**) of the vertical transistor **130** and in contact with the source **142**, a capacitor dielectric layer **150** above and in contact with the first electrode **148**, and a second electrode **152** above and in contact with the capacitor dielectric layer **150**. That is, the memory capacitor **132** may be a vertical capacitor, wherein the first electrode **148**, the capacitor dielectric layer **150** and the second electrode **152** are stacked vertically (i.e., along the Z direction), and the capacitor dielectric layer **150** may be sandwiched between the first electrode **148** and the second electrode **152**. In some implementations, an interconnect structure (e.g., a metal silicide layer) is formed between the drain **144** and the bit line **126** or between the source **142** and the first electrode **148** to reduce contact resistance.

[0037] As shown in FIG. 1A, the second semiconductor structure **118** may further comprise a capacitor contact **154** in contact with a common plate of the second electrode **152** to couple the second electrode **152** of the memory capacitor **132** to the peripheral circuit **106** or directly to ground. In some implementations, an ILD forming the memory capacitor **132** has the same dielectric material (e.g., silicon oxide) as two ILDs extending into the semiconductor pillar **134**. A structure and configuration of the memory capacitor **132** may include any suitable structure and configuration, for example, a planar capacitor, a stack capacitor, a multi-fin capacitor, a cylinder capacitor, a trench capacitor or a substrate-plate capacitor.

[0038] As shown in FIG. 1A, the vertical transistor **130** vertically extends through and is in contact with the gate line, the drain **144** of the vertical transistor **130** at the lower end thereof is in contact with the bit line **126**, and the source **142** of the vertical transistor **130** at the upper end thereof is in contact with the first electrode **148** of the memory capacitor **132**. That is, due to the vertical arrangement of the vertical transistor **130**, the bit line **126** and the memory capacitor **132** can be disposed in different planes in the vertical direction, and coupled to opposite ends of the vertical transistor **130** of the memory cell **128** in the vertical direction. In some implementations, the bit line **126** and the memory capacitor **132** are disposed on the opposite sides of the vertical transistor **130** in the vertical direction, which simplifies routing of the bit line **126** and reduces coupling capacitance between the bit line **126** and the memory capacitor **132** compared with a conventional memory cell in which the bit line and the memory capacitor are disposed on the same side of a planar transistor.

[0039] In some examples, the vertical transistor **130** is vertically disposed between the memory capacitor **132** and the bonding interface **116**. That is, the vertical transistor **130** may be arranged

closer to the peripheral circuit **106** of the first semiconductor structure **102** and the bonding interface **116** than the memory capacitor **132**. Since the bit line **126** and the memory capacitor **132** are coupled to the opposite ends of the vertical transistor **130**, the bit line **126** (as part of the second interconnect layer **124**) is vertically disposed between the vertical transistor **130** and the bonding interface **116** to reduce interconnect routing distance and complexity.

[0040] In some examples, the second semiconductor structure **118** further comprises a second substrate **156** disposed above the memory cell **128**, and a pad-out third interconnect layer **158** above the memory cell **128**. The pad-out third interconnect layer **158** may comprise interconnects (e.g., a contact pad **162**) in one or more ILDs.

[0041] In some examples, the second semiconductor structure **118** further comprises one or more contacts **160** extending through part of the pad-out third interconnect layer **158** and the second substrate **156** to couple the pad-out third interconnect layer **158** to the memory cell **128** and the second interconnect layer **124**. As such, the peripheral circuit **106** may be coupled to the memory cell **128** through the first interconnect layer **110** and the second interconnect layer **124** as well as the second bonding layer **120** and the first bonding layer **112**, and the peripheral circuit **106** and the memory cell **128** may be coupled to an external circuit through the contacts **160** and the pad-out third interconnect layer **158**.

[0042] The second semiconductor structure **118** is illustrated in detail below with reference to FIGS. **1B** to **1D**. FIG. **1B** is an XY cross-sectional structural diagram of a semiconductor device provided by an example, FIG. **1C** is an XZ cross-sectional structural diagram along a C-C cross-section in FIG. **1B**, and FIG. **1D** is an XZ cross-sectional structural diagram along a D-D cross-section in FIG. **1B**.

[0043] As shown in FIG. **1B**, the semiconductor device **100** comprises: a plurality of semiconductor pillars **134** arranged in an array along an X direction and a Y direction; a plurality of gate line groups **166** spaced apart along the X direction, wherein each of the plurality of gate line groups **166** is located between two adjacent ones of the semiconductor pillars **134** along the X direction, and comprises two gate lines **168** spaced apart along the X direction, and each gate line **168** extends along the Y direction and is connected with the plurality of semiconductor pillars **134** arranged along the Y direction; a plurality of shielding structures **170** spaced apart along the X direction and extending along the Y direction, wherein each of the plurality of shielding structures **170** is located between two adjacent ones of the gate line groups **166** along the X direction; and a plurality of bit lines spaced apart along the Y direction, wherein each bit line extends along the X direction and is connected with the plurality of semiconductor pillars **134** arranged along the X direction.

[0044] Here, a material of the shielding structure **170** may include, for example, a metal material. A shielding effect of the shielding structure **170** may be also enhanced by applying a shielding voltage signal to the shielding structure **170**.

[0045] As shown in FIG. **1C**, the semiconductor device **100** further comprises a gate structure **140** on at least a sidewall of the semiconductor pillar **134**, wherein the gate structure **140** comprises a gate dielectric layer **136** and a gate layer **138**, and the gate dielectric layer **136** is in contact with the semiconductor pillar **134**. The semiconductor pillar **134** comprises, along a Z direction, a source **142**, a drain **144** and a channel region **146** between the source **142** and the drain **144**. Each bit line may be connected with the plurality of sources **142** or drains **144** arranged along the X direction; and correspondingly, a memory capacitor may be connected with the drain **144** or the source **142**.

[0046] As shown in FIGS. **1C** and **1D**, the semiconductor device **100** further comprises a bit line gap **172** located between the two adjacent bit lines along the Y direction and extending along the X direction.

[0047] In the above technical solution, the shielding structure is disposed between two adjacent ones of the gate line groups along the X direction, which can reduce the capacitive coupling effect between the adjacent gate line groups, thereby improving the performance of the semiconductor

device. However, the manufacturing process of the shielding structure is difficult and costly, which is unfavorable to size shrinking of the semiconductor device in future.

[0048] In view of this, examples of the present disclosure provide a semiconductor device and a manufacturing method thereof.

[0049] With reference to FIG. 2, FIG. 2 is a flow diagram of a manufacturing method of a semiconductor device provided by examples of the present disclosure. As shown in FIG. 2, examples of the present disclosure provide a manufacturing method of a semiconductor device, comprising: [0050] **S201**: providing a semiconductor structure comprising a semiconductor layer as well as a plurality of first trenches extending along a first direction and a plurality of second trenches extending along a second direction in the semiconductor layer, wherein the first trenches are filled with a first sacrificial layer, and the first direction intersects the second direction; [0051] **S202**: forming an initial gate structure and a second sacrificial layer filling the second trench in the second trench; [0052] **S203**: removing part of the semiconductor layer to expose the first sacrificial layer; [0053] **S204**: etching the first sacrificial layer to form a plurality of third trenches extending along the first direction and arranged along the second direction; [0054] **S205**: removing part of the initial gate structure and the second sacrificial layer in the second trench through the third trench to divide one initial gate structure into two gate structures and to form a first cavity; and [0055] **S206**: forming a first covering layer sealing the third trench to form a second cavity in the third trench, wherein the first cavity and the second cavity are connected.

[0056] In the examples of the present disclosure, the first cavity between the two adjacent gate lines and the second cavity between the two adjacent bit lines are connected, such that the density of the semiconductor device can be increased, the difficulty and cost of manufacturing process of the semiconductor device can be reduced, and the capacitive coupling effect between the adjacent gate lines and between the adjacent bit lines can also be reduced, so as to improve the performance of the semiconductor device and facilitate the size shrinking of the semiconductor device in future.

[0057] In the examples of the present disclosure, the first cavity is disposed between the two adjacent gate lines in the same gate line group, and the second cavity is disposed between the two adjacent bit lines, and the first cavity and the second cavity are connected. A third cavity is further disposed between the two adjacent gate line groups. In a first example of the present disclosure, the first cavity, the second cavity and the third cavity are connected. In a second example of the present disclosure, the first cavity and the second cavity are connected, the first cavity and the third cavity are not connected, and the second cavity and the third cavity are not connected.

[0058] With reference to FIGS. 3A to 3G, FIGS. 3A to 3G are cross-sectional structural diagrams of a semiconductor device provided by a first example in a manufacturing process. The manufacturing process of the semiconductor device provided by the first example of the present disclosure will be illustrated in detail below in conjunction with FIGS. 3A to 3G.

[0059] In the examples of the present disclosure, in **S201**, a semiconductor structure **302** is provided, and comprises a semiconductor layer **304** as well as a plurality of first trenches **306** extending along an X direction and a plurality of second trenches **310** extending along a Y direction in the semiconductor layer **304**, wherein the first trenches **306** are filled with a first sacrificial layer **308**; and the X direction intersects the Y direction.

[0060] In some examples, the semiconductor structure **302** further comprises a plurality of fourth trenches **312** extending along the Y direction, wherein the second trenches **310** and the fourth trenches **312** are arranged alternately along the X direction, the depths of the first trenches **306** along a Z direction are greater than the depths of the fourth trenches **312** along the Z direction, and the depths of the second trenches **310** and the fourth trenches **312** along the Z direction may be the same.

[0061] FIG. 3A is an XY cross-sectional structural diagram of the semiconductor device, and FIG. 3B is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. 3A. As shown in FIGS. 3A and 3B, the semiconductor layer **304** is provided; the semiconductor layer **304** is etched

along the Z direction to form the plurality of first trenches **306** (as shown by a dashed box in FIG. 3A) extending along the X direction and a plurality of semiconductor strips extending along the X direction; a sacrificial material is filled in the first trench **306** to form the first sacrificial layer **308**; the semiconductor strips and the first sacrificial layer **308** are etched along the Z direction to form the plurality of second trenches **310** (as shown by a dotted box in FIG. 3A) and the plurality of fourth trenches **312** (as shown by a dotted box in FIG. 3A) extending along the Y direction, wherein the second trenches **310** and the fourth trenches **312** are arranged alternately along the X direction; and the sacrificial material is filled in the second trench **310** and the fourth trench **312** to form a fourth sacrificial layer **314**.

[0062] As shown in FIG. 3A, the first trench **306** extending along the X direction, the second trench **310** extending along the Y direction and the fourth trench **312** extending along the Y direction jointly divide the semiconductor layer **304** into a plurality of semiconductor pillars **316** arranged in an array along the X direction and the Y direction. In other words, the first sacrificial layer **308** extending along the X direction and the fourth sacrificial layer **314** extending along the Y direction jointly divide the semiconductor layer **304** into the plurality of semiconductor pillars **316** arranged in an array along the X direction and the Y direction.

[0063] Here, the semiconductor layer **304** may comprise at least one elemental semiconductor material (e.g., a silicon (Si) substrate, a germanium (Ge) substrate, etc.), at least one III-V compound semiconductor material (e.g., a gallium nitride (GaN) substrate, a gallium arsenide (GaAs) substrate, an indium phosphide (InP) substrate, etc.), at least one II-VI compound semiconductor material, at least one organic semiconductor material or other semiconductor materials known in the art, and may also comprise other substrates containing a semiconductor material, e.g., a silicon on insulator (SOI) substrate, a germanium on insulator (GeOI) substrate, a polycrystalline semiconductor layer on an insulation layer and a silicon germanium substrate, etc.

[0064] In some examples, the depth of the first trench **306** along the Z direction is greater than the depth of the second trench **310** along the Z direction, and the depth of the first trench **306** along the Z direction is greater than the depth of the fourth trench **312** along the Z direction. On this basis, the height of the first sacrificial layer **308** along the Z direction is greater than the height of the fourth sacrificial layer **314** along the Z direction.

[0065] In some examples, the depth of the second trench **310** along the Z direction and the depth of the fourth trench **312** along the Z direction may be the same.

[0066] Here, a process of forming the first trench **306**, the second trench **310** and the fourth trench **312** by etching may include, but is not limited to, dry etching, wet etching or a combination thereof. For example, the second trench **310** and the fourth trench **312** may be formed using All In One Etch (AIO ET).

[0067] In some examples, materials of the first sacrificial layer **308** and the fourth sacrificial layer **314** may be the same or different.

[0068] Here, the materials of the first sacrificial layer **308** and the fourth sacrificial layer **314** may include, but are not limited to, silicon oxide, silicon nitride or silicon oxynitride, etc.

[0069] Here, a process of forming the first sacrificial layer **308** and the fourth sacrificial layer **314** may include, but is not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD) or atomic layer deposition (ALD), etc.

[0070] Here, a sacrificial material is filled in the first trench **306** to form a first sacrificial material layer filling the first trench **306** and covering a surface of the semiconductor layer **304**; and the first sacrificial material layer is planarized to form the first sacrificial layer **308** exposing the surface of the semiconductor layer **304**. Similarly, a sacrificial material may be filled in the second trench **310** and the fourth trench **312** to form the fourth sacrificial layer **314**.

[0071] Here, the planarization processing may include, but is not limited to, chemical mechanical polishing (CMP).

[0072] Here, the shape of orthographic projection of the semiconductor pillar **316** on an XY plane

may include, but is not limited to, a quadrangle, a circle, or an ellipse, etc. The shape of orthographic projection of the semiconductor pillar **316** on the XY plane is not specifically limited in the present disclosure. FIG. **3A** makes an illustration by taking the shape of orthographic projection of the semiconductor pillar **316** on the XY plane as a quadrangle as an example. The semiconductor pillar **316** comprises two sidewalls oppositely disposed along the X direction and two sidewalls oppositely disposed along the Y direction.

[0073] Here, first the first trench **306** extending along the X direction may be formed by etching, and then the second trench **310** and the fourth trench **312** extending along the Y direction may be formed by etching; alternatively, first the second trench **310** and the fourth trench **312** extending along the Y direction may be formed by etching, and then the first trench **306** extending along the X direction may be formed by etching. In the present disclosure, there are no specific limitations to the etching sequence for forming the first trench **306**, the second trench **310** and the fourth trench **312** by etching.

[0074] FIGS. **3C** to **3G** are XZ cross-sectional structural diagrams along a cross-section parallel to the X direction. As shown in FIG. **3C**, in some examples, the method further comprises: removing the fourth sacrificial layer **314** in the fourth trench **312**; forming a second dielectric layer **318** covering a sidewall and bottom of the fourth trench **312**; filling a sacrificial material in the fourth trench **312** to form a third sacrificial layer **320**; removing part of the third sacrificial layer **320** by back-etching, such that a surface of the third sacrificial layer **320** is lower than the surface of the semiconductor layer **304**; and filling a dielectric material in the fourth trench **312** to form a third dielectric layer **322**.

[0075] In an example, the sacrificial material is filled in the fourth trench **312** to form a third sacrificial material layer filling the fourth trench **312** and covering the surface of the semiconductor layer **304**; the third sacrificial material layer is planarized to form the third sacrificial layer **320** exposing the surface of the semiconductor layer **304**, such that the surface of the third sacrificial layer **320** is substantially flush with the surface of the semiconductor layer **304**; part of the third sacrificial layer **320** is removed by back-etching, such that the surface of the third sacrificial layer **320** is lower than the surface of the semiconductor layer **304**; the dielectric material is filled in the fourth trench **312** to form the third dielectric material layer covering the surface of the semiconductor layer **304**; the third dielectric material layer is planarized to form the third dielectric layer **322** exposing the surface of the semiconductor layer **304**, such that the surface of the third dielectric layer **322** is substantially flush with the surface of the semiconductor layer **304**.

[0076] Here, “substantially flush” may be interpreted as that after chemical mechanical polishing processing, the surface of the formed third dielectric layer **322** and the surface of the semiconductor layer **304** are flush on the same plane, and that during the chemical mechanical polishing process, the surface of the formed third dielectric layer **322** and the surface of the semiconductor layer **304** are not flush on the same plane within the error range of the process.

[0077] Here, the second dielectric layer **318** covers the bottom and sidewall of the third sacrificial layer **320**, and the third dielectric layer **322** covers the top of the third sacrificial layer **320**. The third sacrificial layer **320** will be removed during a subsequent process to form the third cavity, so as to reduce the capacitive coupling effect between two adjacent gate line groups (or two gate lines).

[0078] In some examples, materials of the second dielectric layer **318** and the third dielectric layer **322** may be the same or different.

[0079] Here, the materials of the second dielectric layer **318** and the third dielectric layer **322** may include, but are not limited to, silicon oxide.

[0080] Here, a material of the third sacrificial layer **320** may include, but is not limited to, silicon oxide, silicon nitride or silicon oxynitride, etc.

[0081] Here, a process of forming the second dielectric layer **318**, the third dielectric layer **322** and the third sacrificial layer **320** may include, but is not limited to, CVD, PVD or ALD, etc.

[0082] In the examples of the present disclosure, in S202, an initial gate structure 326 and a second sacrificial layer 328 filling the second trench 310 are formed in the second trench 310.

[0083] As shown in FIG. 3D, the fourth sacrificial layer 314 in the second trench 310 is removed. In some examples, S202 comprises: forming a first dielectric layer 324 covering a sidewall and bottom of the second trench 310; forming the initial gate structure 326 covering the sidewall and the bottom of the second trench 310; filling a sacrificial material in the second trench 310 to form the second sacrificial layer 328; back-etching part of the initial gate structure 326 and the second sacrificial layer 328, such that surfaces of the initial gate structure 326 and the second sacrificial layer 328 are lower than the surface of the semiconductor layer 304; and filling a dielectric material in the second trench 310 to form a fourth dielectric layer 330.

[0084] In an example, the initial gate structure 326 covering the sidewall and the bottom of the second trench 310 is formed; the sacrificial material is filled in the second trench 310 to form the second sacrificial layer 328, such that a surface of the initial gate structure 326, a surface of the second sacrificial layer 328 and the surface of the semiconductor layer 304 are substantially flush; part of the initial gate structure 326 and the second sacrificial layer 328 is removed by back-etching, such that the surfaces of both the initial gate structure 326 and the second sacrificial layer 328 are lower than the surface of the semiconductor layer 304; the dielectric material is filled in the second trench 310 to form a fourth dielectric material layer covering the surface of the semiconductor layer 304; the fourth dielectric material layer is planarized to form the fourth dielectric layer 330 exposing the surface of the semiconductor layer 304, such that a surface of the fourth dielectric layer 330 is substantially flush with the surface of the semiconductor layer 304.

[0085] Here, during a front side process of the semiconductor layer, the second sacrificial layer 328 may be buried in the second trench 310, and the third sacrificial layer 320 may be buried in the fourth trench 312. Subsequently during a back side process of the semiconductor layer, the second sacrificial layer 328 in the second trench 310 may be removed to form the first cavity, and the third sacrificial layer 320 in the fourth trench 312 may be removed to form the third cavity. In addition, the first cavity, the third cavity, and the second cavity between the subsequent bit lines may be also connected, and the first cavity, the third cavity and the second cavity are formed integrally.

[0086] Here, “substantially flush” may be interpreted as that after chemical mechanical polishing processing, the surface of the formed fourth dielectric layer 330 and the surface of the semiconductor layer 304 are flush on the same plane, and that during the chemical mechanical polishing process, the surface of the formed fourth dielectric layer 330 and the surface of the semiconductor layer 304 are not flush on the same plane within the error range of the process.

[0087] Here, the first dielectric layer 324 (or the initial gate structure 326) covers the bottom and sidewall of the second sacrificial layer 328, and the fourth dielectric layer 330 covers the top of the second sacrificial layer 328. The second sacrificial layer 328 will be removed during a subsequent process to form the first cavity, so as to reduce the capacitive coupling effect between the two adjacent gate lines.

[0088] In some examples, materials of the first dielectric layer 324 and the fourth dielectric layer 330 may be the same or different.

[0089] Here, the materials of the first dielectric layer 324 and the fourth dielectric layer 330 may include, but are not limited to, silicon oxide.

[0090] Here, a material of the initial gate structure 326 may comprise a conductive material, e.g., a metal material. In an example, the material of the initial gate structure 326 may include tungsten.

[0091] In the examples of the present disclosure, the materials of the second sacrificial layer 328 and the third sacrificial layer 320 or be the same or different.

[0092] Here, a process of forming the first dielectric layer 324, the fourth dielectric layer 330, the second sacrificial layer 328 and the initial gate structure 326 may include, but is not limited to, CVD, PVD or ALD, etc.

[0093] In the examples of the present disclosure, in S203, part of the semiconductor layer 304 is

removed to expose the first sacrificial layer **308**.

[0094] As shown in FIG. **3E**, the back side of the semiconductor layer **304** is thinned to expose the first sacrificial layer **308**. Since the height of the first sacrificial layer **308** along the Z direction is greater than the height of the fourth sacrificial layer **314** along the Z direction, the remaining semiconductor layer forms the bit lines **356** while removing part of the semiconductor layer **304** to expose the first sacrificial layer **308**. The plurality of bit lines **356** are spaced apart along the Y direction, and the bit lines **356** extending along the X direction are connected with the plurality of semiconductor pillars **316** spaced apart along the X direction.

[0095] Here, the semiconductor pillar **316** comprises, along the Z direction, a source, a drain and a channel region between the source and the drain, wherein each bit line may be connected with the plurality of sources or drains arranged along the X direction; and correspondingly, the memory capacitor may be connected with the source or the drain.

[0096] In the examples of the present disclosure, in **S204**, the first sacrificial layer **308** is etched to form the plurality of third trenches extending along the X direction and arranged along the Y direction.

[0097] Here, part of the first sacrificial layer **308** may be removed from the back side of the semiconductor layer **304** to form the plurality of third trenches extending along the X direction and arranged along the Y direction, wherein the third trench is located between two adjacent bit lines **356** along the Y direction.

[0098] In the examples of the present disclosure, in **S205**, part of the initial gate structure **326** and the second sacrificial layer **328** in the second trench **310** are removed through the third trench to divide one initial gate structure **326** into two gate structures **332** and to form the first cavity **334**.

[0099] As shown in FIG. **3F**, the first dielectric layer **324** covering the bottom of the second trench **310** is removed through the third trench to expose the initial gate structure **326** covering the bottom of the second trench **310**. At this point, the first dielectric layer **324** covering the two oppositely disposed sidewalls of the second trench **310** forms a gate dielectric layer **342**. The initial gate structure **326** covering the bottom of the second trench **310** is removed through the third trench to divide one initial gate structure **326** into two gate layers **344**, that is, the initial gate structure **326** covering the two oppositely disposed sidewalls of the second trench **310** forms the gate layers **344**, wherein the gate dielectric layer **342** covering the sidewall of the semiconductor pillar **316** and the gate layer **344** covering the gate dielectric layer **342** jointly form the gate structure **332**. In addition, the second sacrificial layer **328** in the second trench **310** is removed through the third trench to form the first cavity **334**.

[0100] Here, the gate structure **332** is located on a sidewall of the semiconductor pillar **316**, and the gate structures **332** of the plurality of semiconductor pillars arranged along the Y direction are connected to form a gate line. Two gate lines in the same second trench **310** form a gate line group that is located between two adjacent semiconductor pillars **316** along the X direction, and each gate line group comprises two gate lines spaced apart along the X direction and extending along the Y direction.

[0101] Here, the first cavity **334** is located between the two gate lines in the same gate line group, and the first cavity **334** extends along the Y direction.

[0102] In some examples, at the same time as performing **S205**, the method further comprises: removing the third sacrificial layer **320** in the fourth trench **312** through the third trench to form the third cavity **336** in the fourth trench **312**.

[0103] As shown in FIG. **3F**, the second dielectric layer **318** covering the bottom of the fourth trench **312** is removed through the third trench to expose the third sacrificial layer **320** covering the bottom of the fourth trench **312**; and the third sacrificial layer **320** in the fourth trench **312** is removed through the third trench to form the third cavity **336**.

[0104] Here, the third cavity **336** is located between the two adjacent semiconductor pillars **316** along the X direction, or the third cavity **336** is located between the two adjacent gate line groups

along the X direction, and the third cavity **336** extends along the Y direction.

[0105] Here, the third cavity **336** has two surfaces disposed oppositely along the Z direction. Since the third cavity **336** is formed by removing the third sacrificial layer **320**, the surface of the third cavity **336** away from the bit line **356** is a flat surface, that is, a surface of the third dielectric layer **322** close to the third cavity **336** is a flat surface.

[0106] In the examples of the present disclosure, in **S206**, a first covering layer **338** sealing the third trench is formed to form a second cavity **340** in the third trench, wherein the first cavity **334**, the second cavity **340** and the third cavity **336** are connected.

[0107] As shown in FIG. **3G**, the first covering layer **338** sealing the third trench is formed to form the second cavity **340** in the third trench, wherein the first cavity **334**, the second cavity **340** and the third cavity **336** are connected.

[0108] Here, the second cavity **340** is located between the two adjacent bit lines **356** along the Y direction and extends along the X direction.

[0109] Here, a material of the first covering layer **338** may include, but is not limited to, an insulation material, e.g., silicon oxide.

[0110] Here, the second cavity **340** has two surfaces disposed oppositely along the Z direction. Since the first cavity **334**, the second cavity **340** and the third cavity **336** are sealed by forming the first covering layer **338**, the surface of the second cavity **340** away from the third cavity **336** has a convex shape towards a direction away from the third cavity **336**.

[0111] In the first example of the present disclosure, the first cavity, the second cavity and the third cavity are connected. As such, not only can the manufacturing process difficulty and cost of the semiconductor device be reduced, but also the capacitive coupling effect between the adjacent gate lines and between the adjacent bit lines can be reduced, so as to improve the performance of the semiconductor device and facilitate the size shrinking of the semiconductor device in future.

[0112] With reference to FIGS. **4A** to **4G**, FIGS. **4A** to **4G** are cross-sectional structural diagrams of a semiconductor device provided by a second example in a manufacturing process. The manufacturing process of the semiconductor device provided by the second example of the present disclosure will be illustrated in detail below in conjunction with FIGS. **4A** to **4G**.

[0113] FIG. **4A** is an XY cross-sectional structural diagram of the semiconductor device, and FIG. **4B** is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. **4A**. As shown in FIGS. **4A** and **4B**, a semiconductor structure **402** is provided and comprises a semiconductor layer **404**; a plurality of first trenches **406** located in the semiconductor layer **404** and extending along an X direction; a first sacrificial layer **408** in the first trenches **406**; a plurality of second trenches **410** and a plurality of fourth trenches **412** located in the semiconductor layer **404** and extending along a Y direction, wherein the second trenches **410** and the fourth trenches **412** are arranged alternately along the X direction; and a fourth sacrificial layer **414** in the second trenches **410** and the fourth trenches **412**.

[0114] As shown in FIG. **4A**, the first trench **406** extending along the X direction, the second trench **410** extending along the Y direction and the fourth trench **412** extending along the Y direction jointly divide the semiconductor layer **404** into a plurality of semiconductor pillars **416** arranged in an array along the X direction and the Y direction. In other words, the first sacrificial layer **408** extending along the X direction and the fourth sacrificial layer **414** extending along the Y direction jointly divide the semiconductor layer **404** into the plurality of semiconductor pillars **416** arranged in an array along the X direction and the Y direction.

[0115] FIGS. **4C** to **4G** are XZ cross-sectional structural diagrams along a cross-section parallel to the X direction. As shown in FIG. **4C**, in some examples, the method further comprises: removing the fourth sacrificial layer **414** in the fourth trench **412**; forming a second dielectric layer **418** covering a sidewall and bottom of the fourth trench **412**; forming a blocking layer **420** covering the sidewall and the bottom of the fourth trench **412**, wherein a surface of the blocking layer **420** is substantially flush with a surface of the semiconductor layer **404**; removing part of the blocking

layer **420** by back-etching, such that the surface of the blocking layer **420** is lower than the surface of the semiconductor layer **404**; and forming a second covering layer **422** sealing the fourth trench **412** to form a third cavity **436** in the fourth trench **412**.

[0116] Here, the second dielectric layer **418** covers the bottom and sidewall of the fourth trench **412**; the blocking layer **420** covers the bottom and sidewall of the second dielectric layer **418**, and covers the bottom and sidewall of the third cavity **436**; and the second covering layer **422** covers the top of the third cavity **436**. That is, the blocking layer **420** and the second covering layer **422** jointly surround and form the third cavity **436**.

[0117] Here, the third cavity **436** has two surfaces disposed oppositely along the Z direction. Since the third cavity **436** is sealed by forming the second covering layer **422**, the surface of the third cavity **436** away from the semiconductor layer **404** has a convex shape towards a direction away from the semiconductor layer **404**.

[0118] Here, a material of the second dielectric layer **418** may include, but is not limited to, silicon oxide.

[0119] Here, in the present disclosure, there are no specific limitations to a material of the blocking layer **420** as long as the blocking layer **420** can be prevented from being removed to damage the morphology of the third cavity **436** during a backside process of the semiconductor layer.

[0120] Here, a material of the second covering layer **422** may include, but is not limited to, an insulation material, e.g., silicon oxide.

[0121] Here, a process of forming the second dielectric layer **418**, the blocking layer **420** and the second covering layer **422** may include, but is not limited to, CVD, PVD or ALD, etc.

[0122] In the examples of the present disclosure, in **S202**, an initial gate structure **426** and a second sacrificial layer **428** filling the second trench **410** are formed in the second trench **410**.

[0123] As shown in FIG. **4D**, the fourth sacrificial layer **414** in the second trench **410** is removed. In some examples, in **S202**, the method further comprises: forming a first dielectric layer **424** covering a sidewall and bottom of the second trench **410**; forming the initial gate structure **426** covering the sidewall and the bottom of the second trench **410**; filling a sacrificial material in the second trench **410** to form the second sacrificial layer **428**, wherein a surface of the initial gate structure **426**, a surface of the second sacrificial layer **428** and a surface of the semiconductor layer **404** are substantially flush; removing part of the initial gate structure **426** and the second sacrificial layer **428** by back-etching, such that the surfaces of the initial gate structure **426** and the second sacrificial layer **428** are lower than the surface of the semiconductor layer **404**; filling a dielectric material in the second trench **410** to form a fourth dielectric material layer covering the surface of the semiconductor layer **404**; and planarizing the fourth dielectric material layer to form a fourth dielectric layer **430** exposing the surface of the semiconductor layer **404**, wherein a surface of the fourth dielectric layer **430** is substantially flush with the surface of the semiconductor layer **404**.

[0124] Here, the first dielectric layer **424** and the initial gate structure **426** cover the bottom and sidewall of the second sacrificial layer **428**, and the fourth dielectric layer **430** covers the top of the second sacrificial layer **428**. The second sacrificial layer **428** will be removed during a subsequent process to form the first cavity, so as to reduce the capacitive coupling effect between the two adjacent gate lines.

[0125] In the examples of the present disclosure, in **S203**, part of the semiconductor layer **404** is removed to expose the first sacrificial layer **408**.

[0126] As shown in FIG. **4E**, the back side of the semiconductor layer **404** is thinned to expose the first sacrificial layer **408**. Since the height of the first sacrificial layer **408** along the Z direction is greater than the height of the fourth sacrificial layer **414** along the Z direction, the remaining semiconductor layer forms the bit lines **456** while removing part of the semiconductor layer **404** to expose the first sacrificial layer **408**. The plurality of bit lines **456** are spaced apart along the Y direction, and the bit lines **456** extending along the X direction are connected with the plurality of semiconductor pillars **416** spaced apart along the X direction.

[0127] Here, the semiconductor pillar **416** comprises, along the Z direction, a source, a drain and a channel region between the source and the drain, wherein each bit line may be connected with the plurality of sources or drains arranged along the X direction; and correspondingly, the memory capacitor may be connected with the source or the drain.

[0128] In the examples of the present disclosure, in **S204**, the first sacrificial layer **408** is etched to form a plurality of third trenches extending along the X direction and arranged along the Y direction.

[0129] Here, part of the first sacrificial layer **408** may be removed from the back side of the semiconductor layer **404** to form the plurality of third trenches extending along the X direction and arranged along the Y direction, wherein the third trench is located between the two adjacent bit lines **456** along the Y direction.

[0130] In the examples of the present disclosure, in **S205**, part of the initial gate structure **426** and the second sacrificial layer **428** in the second trench **410** are removed through the third trench to divide one initial gate structure **426** into two gate structures **432** and to form the first cavity **434**.

[0131] As shown in FIG. **4F**, the first dielectric layer **424** covering the bottom of the second trench **410** is removed through the third trench to expose the initial gate structure **426** covering the bottom of the second trench **410**. At this point, the first dielectric layer **424** covering the two oppositely disposed sidewalls of the second trench **410** forms a gate dielectric layer **442**. The initial gate structure **426** covering the bottom of the second trench **410** is removed through the third trench to divide one initial gate structure **426** into two gate layers **444**, that is, the initial gate structure **426** covering the two oppositely disposed sidewalls of the second trench **410** forms the gate layers **444**, wherein the gate dielectric layer **442** covering the sidewall of the semiconductor pillar **416** and the gate layer **444** covering the gate dielectric layer **442** jointly form the gate structure **432**. In addition, the second sacrificial layer **428** in the second trench **410** is removed through the third trench to form the first cavity **434**.

[0132] Here, the gate structure **432** is located on a sidewall of the semiconductor pillar **416**, and the gate structures **432** of the plurality of semiconductor pillars arranged along the Y direction are connected to form a gate line. Two gate lines in the same second trench **410** form a gate line group that is located between the two adjacent semiconductor pillars **416** along the X direction, and each gate line group comprises two gate lines spaced apart along the X direction and extending along the Y direction.

[0133] Here, the first cavity **434** is located between the two gate lines in the same gate line group, and the first cavity **434** extends along the Y direction.

[0134] Here, the blocking layer **420** in the fourth trench **412** may prevent a process of forming the first cavity **434** from damaging the morphology of the third cavity **436**. As such, the blocking layer **420** may serve as an etch stop layer.

[0135] In the examples of the present disclosure, in **S206**, a first covering layer **438** sealing the third trench is formed to form a second cavity **440** in the third trench, wherein the first cavity **434** and the second cavity **440** are connected.

[0136] As shown in FIG. **4G**, the first covering layer **438** sealing the third trench is formed to form the second cavity **440** in the third trench, wherein the first cavity **434** and the second cavity **440** are connected, the first cavity **434** and the third cavity **436** are not connected, and the second cavity **440** and the third cavity **436** are not connected.

[0137] Here, the second cavity **440** is located between the two adjacent bit lines **456** along the Y direction and extends along the X direction.

[0138] Here, a material of the first covering layer **438** may include, but is not limited to, an insulation material, e.g., silicon oxide.

[0139] Here, the second cavity **440** has two surfaces disposed oppositely along the Z direction. Since the first cavity **434** and the second cavity **440** are sealed by forming the first covering layer **438**, the surface of the second cavity **440** away from the first cavity **434** has a convex shape

towards a direction away from the first cavity **434**.

[0140] In the second example of the present disclosure, the first cavity and the second cavity are connected, the first cavity and the third cavity are not connected, and the second cavity and the third cavity are not connected. As such, not only can the manufacturing process difficulty and cost of the semiconductor device be reduced, but also the capacitive coupling effect between the adjacent gate lines and between the adjacent bit lines can be reduced, so as to improve the performance of the semiconductor device and facilitate the size shrinking of the semiconductor device in future.

[0141] In the examples of the present disclosure, the second cavity is disposed between the two adjacent bit lines, and the third cavity is disposed between the two adjacent gate line groups. In the third example of the present disclosure, the second cavity and the third cavity are connected. In the fourth example of the present disclosure, the second cavity and the third cavity are not connected.

[0142] With reference to FIGS. 5A to 5G, FIGS. 5A to 5G are cross-sectional structural diagrams of a semiconductor device provided by a third example in a manufacturing process. The manufacturing process of the semiconductor device provided by the third example of the present disclosure will be illustrated in detail below in conjunction with FIGS. 5A to 5G.

[0143] FIG. 5A is an XY cross-sectional structural diagram of the semiconductor device, and FIG. 5B is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. 5A. As shown in FIGS. 5A and 5B, a semiconductor structure **502** is provided and comprises a semiconductor layer **504**; a plurality of first trenches **506** located in the semiconductor layer **504** and extending along an X direction; a first sacrificial layer **508** in the first trenches **506**; a plurality of second trenches **510** and a plurality of fourth trenches **512** located in the semiconductor layer **504** and extending along a Y direction, wherein the second trenches **510** and the fourth trenches **512** are arranged alternately along the X direction; and a fourth sacrificial layer **514** in the second trenches **510** and the fourth trenches **512**.

[0144] As shown in FIG. 5A, the first trench **506** extending along the X direction, the second trench **510** extending along the Y direction and the fourth trench **512** extending along the Y direction jointly divide the semiconductor layer **504** into a plurality of semiconductor pillars **516** arranged in an array along the X direction and the Y direction. In other words, the first sacrificial layer **508** extending along the X direction and the fourth sacrificial layer **514** extending along the Y direction jointly divide the semiconductor layer **504** into the plurality of semiconductor pillars **516** arranged in an array along the X direction and the Y direction.

[0145] FIGS. 5C to 5G are XZ cross-sectional structural diagrams along a cross-section parallel to the X direction. As shown in FIG. 5C, in some examples, the method further comprises: removing a fourth sacrificial layer **514** in the fourth trench **512**; forming a second dielectric layer **518** covering a sidewall and bottom of the fourth trench **512**; filling a sacrificial material in the fourth trench **512** to form a third sacrificial layer **520**, wherein a surface of the third sacrificial layer **520** is substantially flush with a surface of the semiconductor layer **504**; removing part of the third sacrificial layer **520** by back-etching, such that the surface of the third sacrificial layer **520** is lower than the surface of the semiconductor layer **504**; filling a dielectric material in the fourth trench **512** to form a third dielectric material layer covering the surface of the semiconductor layer **504**; and planarizing the third dielectric material layer to form a third dielectric layer **522** exposing the surface of the semiconductor layer **504**, wherein the surface of the third dielectric layer **522** is substantially flush with the surface of the semiconductor layer **504**.

[0146] Here, the second dielectric layer **518** covers the bottom and sidewall of the third sacrificial layer **520**, and the third dielectric layer **522** covers the top of the third sacrificial layer **520**. The third sacrificial layer **520** will be removed during a subsequent process to form the third cavity, so as to reduce the capacitive coupling effect between two adjacent gate line groups (or two adjacent gate lines).

[0147] As shown in FIG. 5D, the fourth sacrificial layer **514** in the second trench **510** is removed; a

first dielectric layer **524** covering a sidewall and bottom of the second trench **510** is formed; an initial gate structure **526** covering the sidewall and the bottom of the second trench **510** is formed; a dielectric material is filled in the second trench **510** to form a fifth dielectric layer **528**, such that a surface of the initial gate structure **526**, a surface of the fifth dielectric layer **528** and the surface of the semiconductor layer **504** are substantially flush; part of the initial gate structure **526** and the fifth dielectric layer **528** are removed by back-etching, such that the surfaces of the initial gate structure **526** and the fifth dielectric layer **528** are lower than the surface of the semiconductor layer **504**; a dielectric material is filled in the second trench **510** to form a sixth dielectric material layer covering the surface of the semiconductor layer **504**; the sixth dielectric material layer is planarized to form a sixth dielectric layer **530** exposing the surface of the semiconductor layer **504**, such that a surface of the sixth dielectric layer **530** is substantially flush with the surface of the semiconductor layer **504**.

[0148] Here, the first dielectric layer **524** (or the initial gate structure **526**) covers the bottom and sidewall of the fifth dielectric layer **528**, and the sixth dielectric layer **530** covers the top of the fifth dielectric layer **528**.

[0149] As shown in FIG. 5E, the back side of the semiconductor layer **504** is thinned to expose the first sacrificial layer **508**. Since the height of the first sacrificial layer **508** along the Z direction is greater than the height of the fourth sacrificial layer **514** along the Z direction, the remaining semiconductor layer forms the bit lines **556** while removing part of the semiconductor layer **504** to expose the first sacrificial layer **508**. The plurality of bit lines **556** are spaced apart along the Y direction, and the bit lines **556** extending along the X direction are connected with the plurality of semiconductor pillars **516** spaced apart along the X direction.

[0150] Here, the semiconductor pillar **516** comprises, along the Z direction, a source, a drain and a channel region between the source and the drain, wherein each bit line may be connected with the plurality of sources or drains arranged along the X direction; and correspondingly, the memory capacitor may be connected with the source or the drain.

[0151] Here, part of the first sacrificial layer **508** may be removed from the back side of the semiconductor layer **504** to form the plurality of third trenches extending along the X direction and arranged along the Y direction, wherein the third trench is located between the two adjacent bit lines **556** along the Y direction.

[0152] As shown in FIG. 5F, the first dielectric layer **524** covering the bottom of the second trench **510** is removed through the third trench to expose the initial gate structure **526** covering the bottom of the second trench **510**. At this point, the first dielectric layer **524** covering the two oppositely disposed sidewalls of the second trench **510** forms a gate dielectric layer **542**. The initial gate structure **526** covering the bottom of the second trench **510** is removed through the third trench to divide one initial gate structure **526** into two gate layers **544**, that is, the initial gate structure **526** covering the two oppositely disposed sidewalls of the second trench **510** forms the gate layers **544**, wherein the gate dielectric layer **542** covering the sidewall of the semiconductor pillar **516** and the gate layer **544** covering the gate dielectric layer **542** jointly form the gate structure **532**.

[0153] Here, the gate structure **532** is located on a sidewall of the semiconductor pillar **516**, and the gate structures **532** of the plurality of semiconductor pillars arranged along the Y direction are connected to form a gate line. Two gate lines in the same second trench **510** form a gate line group that is located between the two adjacent semiconductor pillars **516** along the X direction, and each gate line group comprises two gate lines spaced apart along the X direction and extending along the Y direction, and the two gate lines are isolated by the fifth dielectric layer **528**.

[0154] As shown in FIG. 5F, the second dielectric layer **518** covering the bottom of the fourth trench **512** is removed through the third trench to expose the third sacrificial layer **520** covering the bottom of the fourth trench **512**; and the third sacrificial layer **520** in the fourth trench **512** is removed through the third trench to form the third cavity **536**.

[0155] Here, the third cavity **536** is located between the two adjacent semiconductor pillars **516**

along the X direction, or the third cavity **536** is located between the two adjacent gate line groups along the X direction and extends along the Y direction.

[0156] Here, the third cavity **536** has two surfaces disposed oppositely along the Z direction. Since the third cavity **536** is formed by removing the third sacrificial layer **520**, the surface of the third cavity **536** away from the bit line **556** is a flat surface, that is, a surface of the third dielectric layer **522** close to the third cavity **536** is a flat surface.

[0157] As shown in FIG. **5G**, a first covering layer **538** sealing the third trench is formed to form a second cavity **540** in the third trench, wherein the second cavity **540** and the third cavity **536** are connected.

[0158] Here, the second cavity **540** is located between the two adjacent bit lines **556** along the Y direction and extends along the X direction.

[0159] Here, the second cavity **540** has two surfaces disposed oppositely along the Z direction. Since the second cavity **540** and the third cavity **536** are sealed by forming the first covering layer **538**, the surface of the second cavity **540** away from the third cavity **536** has a convex shape towards a direction away from the third cavity **536**.

[0160] In the third example of the present disclosure, the second cavity and the third cavity are connected. As such, not only can the manufacturing process difficulty and cost of the semiconductor device be reduced, but also the capacitive coupling effect between the adjacent gate line groups and between the adjacent bit lines can be reduced, so as to improve the performance of the semiconductor device and facilitate the size shrinking of the semiconductor device in future.

[0161] With reference to FIGS. **6A** to **6G**, FIGS. **6A** to **6G** are cross-sectional structural diagrams of a semiconductor device provided by a fourth example in a manufacturing process. The manufacturing process of the semiconductor device provided by the fourth example of the present disclosure will be illustrated in detail below in conjunction with FIGS. **6A** to **6G**.

[0162] FIG. **6A** is an XY cross-sectional structural diagram of the semiconductor device, and FIG. **6B** is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. **6A**. As shown in FIGS. **6A** and **6B**, a semiconductor structure **602** is provided and comprises a semiconductor layer **604**; a plurality of first trenches **606** located in the semiconductor layer **604** and extending along an X direction; a first sacrificial layer **608** in the first trenches **606**; a plurality of second trenches **610** and a plurality of fourth trenches **612** located in the semiconductor layer **604** and extending along a Y direction, wherein the second trenches **610** and the fourth trenches **612** are arranged alternately along the X direction; and a fourth sacrificial layer **614** in the second trenches **610** and the fourth trenches **612**.

[0163] As shown in FIG. **6A**, the first trench **606** extending along the X direction, the second trench **610** extending along the Y direction and the fourth trench **612** extending along the Y direction jointly divide the semiconductor layer **604** into a plurality of semiconductor pillars **616** arranged in an array along the X direction and the Y direction. In other words, the first sacrificial layer **608** extending along the X direction and the fourth sacrificial layer **614** extending along the Y direction jointly divide the semiconductor layer **604** into the plurality of semiconductor pillars **616** arranged in an array along the X direction and the Y direction.

[0164] FIGS. **6C** to **6G** are XZ cross-sectional structural diagrams along a cross-section parallel to the X direction. As shown in FIG. **6C**, the fourth sacrificial layer **614** in the fourth trench **612** is removed; a second dielectric layer **618** covering a sidewall and bottom of the fourth trench **612** is formed; a blocking layer **620** covering the sidewall and the bottom of the fourth trench **612** is formed, wherein a surface of the blocking layer **620** is substantially flush with a surface of the semiconductor layer **604**; part of the blocking layer **620** is removed by back-etching, such that the surface of the blocking layer **620** is lower than the surface of the semiconductor layer **604**; and a second covering layer **622** sealing the fourth trench **612** is formed to form a third cavity **636** in the fourth trench **612**.

[0165] Here, the second dielectric layer **618** covers the bottom and sidewall of the fourth trench

612; the blocking layer **620** covers the bottom and sidewall of the second dielectric layer **618**, and covers the bottom and sidewall of the third cavity **636**; and the second covering layer **622** covers the top of the third cavity **636**. That is, the blocking layer **620** and the second covering layer **622** jointly surround and form the third cavity **636**.

[0166] Here, the third cavity **636** has two surfaces disposed oppositely along the Z direction. Since the third cavity **636** is sealed by forming the second covering layer **622**, the surface of the third cavity **636** away from the semiconductor layer **604** has a convex shape towards a direction away from the semiconductor layer **604**.

[0167] As shown in FIG. 6D, the fourth sacrificial layer **614** in the second trench **610** is removed; a first dielectric layer **624** covering a sidewall and bottom of the second trench **610** is formed; an initial gate structure **626** covering the sidewall and the bottom of the second trench **610** is formed; a dielectric material is filled in the second trench **610** to form a fifth dielectric layer **628**, such that a surface of the initial gate structure **626**, a surface of the fifth dielectric layer **628** and the surface of the semiconductor layer **604** are substantially flush; part of the initial gate structure **626** and the fifth dielectric layer **628** are removed by back-etching, such that the surfaces of the initial gate structure **626** and the fifth dielectric layer **628** are lower than the surface of the semiconductor layer **604**; a dielectric material is filled in the second trench **610** to form a sixth dielectric material layer covering the surface of the semiconductor layer **604**; the sixth dielectric material layer is planarized to form a sixth dielectric layer **630** exposing the surface of the semiconductor layer **604**, such that a surface of the sixth dielectric layer **630** is substantially flush with the surface of the semiconductor layer **604**.

[0168] Here, the first dielectric layer **624** and the initial gate structure **626** cover the bottom and sidewall of the fifth dielectric layer **628**, and the sixth dielectric layer **630** covers the top of the fifth dielectric layer **628**.

[0169] As shown in FIG. 6E, the back side of the semiconductor layer **604** is thinned to expose the first sacrificial layer **608**. Since the height of the first sacrificial layer **608** along the Z direction is greater than the height of the fourth sacrificial layer **614** along the Z direction, the remaining semiconductor layer forms the bit lines **656** while removing part of the semiconductor layer **604** to expose the first sacrificial layer **608**. The plurality of bit lines **656** are spaced apart along the Y direction, and the bit lines **656** extending along the X direction are connected with the plurality of semiconductor pillars **616** spaced apart along the X direction.

[0170] Here, the semiconductor pillar **616** comprises, along the Z direction, a source, a drain and a channel region between the source and the drain, wherein each bit line may be connected with the plurality of sources or drains arranged along the X direction; and correspondingly, the memory capacitor may be connected with the source or the drain.

[0171] Here, part of the first sacrificial layer **608** may be removed from the back side of the semiconductor layer **604** to form the plurality of third trenches extending along the X direction and arranged along the Y direction, wherein the third trench is located between the two adjacent bit lines **656** along the Y direction.

[0172] As shown in FIG. 6F, the first dielectric layer **624** covering the bottom of the second trench **610** is removed through the third trench to expose the initial gate structure **626** covering the bottom of the second trench **610**. At this point, the first dielectric layer **624** covering the two oppositely disposed sidewalls of the second trench **610** forms a gate dielectric layer **642**. The initial gate structure **626** covering the bottom of the second trench **610** is removed through the third trench to divide one initial gate structure **626** into two gate layers **644**, that is, the initial gate structure **626** covering the two oppositely disposed sidewalls of the second trench **610** forms the gate layers **644**, wherein the gate dielectric layer **642** covering the sidewall of the semiconductor pillar **616** and the gate layer **644** covering the gate dielectric layer **642** jointly form the gate structure **632**.

[0173] Here, the gate structure **632** is located on a sidewall of the semiconductor pillar **616**, and the gate structures **632** of the plurality of semiconductor pillars arranged along the Y direction are

connected to form a gate line. Two gate lines in the same second trench **610** form a gate line group that is located between the two adjacent semiconductor pillars **616** along the X direction, each gate line group comprises two gate lines spaced apart along the X direction and extending along the Y direction, and the two gate lines are isolated by the fifth dielectric layer **628**.

[0174] Here, the blocking layer **620** in the fourth trench **612** may prevent a process of forming the gate structure **632** from damaging the morphology of the third cavity **636**. As such, the blocking layer **620** may serve as an etch stop layer.

[0175] As shown in FIG. **6G**, a first covering layer **638** sealing the third trench is formed to form a second cavity **640** in the third trench, wherein the second cavity **640** and the third cavity **636** are not connected.

[0176] Here, the second cavity **640** is located between the two adjacent bit lines **656** along the Y direction and extends along the X direction.

[0177] Here, the second cavity **640** has two surfaces disposed oppositely along the Z direction. Since the second cavity **640** is sealed by forming the first covering layer **638**, the surface of the second cavity **640** away from the third cavity **636** has a convex shape towards a direction away from the third cavity **636**.

[0178] In the fourth example of the present disclosure, the second cavity and the third cavity are not connected. As such, not only can the manufacturing process difficulty and cost of the semiconductor device be reduced, but also the capacitive coupling effect between the adjacent gate line groups and between the adjacent bit lines can be reduced, so as to improve the performance of the semiconductor device and facilitate the size shrinking of the semiconductor device in future.

[0179] With reference to FIGS. **7A** to **7C**, FIGS. **7A** to **7C** are cross-sectional structural diagrams of a semiconductor device provided by a first example. FIG. **7A** is an XY cross-sectional structural diagram of a semiconductor device provided by a first example, FIG. **7B** is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. **7A**, and FIG. **7C** is an XZ cross-sectional structural diagram along a C-C cross-section in FIG. **7A**.

[0180] As shown in FIGS. **7A** to **7C**, examples of the present disclosure provide a semiconductor device. The semiconductor device **300** comprises: a plurality of semiconductor pillars **316** arranged along an X direction and a Y direction, wherein the X direction intersects the Y direction; a plurality of gate lines **346** spaced apart along the X direction, wherein each gate line **346** extends along the Y direction and is connected with the plurality of semiconductor pillars **316** arranged along the Y direction; a first cavity **334** located between the two adjacent ones of the plurality of gate lines **346** along the X direction and extending along the Y direction; a plurality of bit lines **356** spaced apart along the Y direction, wherein each bit line **356** extends along the X direction and is connected with the plurality of semiconductor pillars **316** arranged along the X direction; and a second cavity **340** located between two adjacent ones of the plurality of bit lines **356** along the Y direction and extending along the X direction, wherein the second cavity **340** and the first cavity **334** are connected.

[0181] As shown in FIG. **7A**, in some examples, the plurality of gate lines **346** comprises a plurality of gate line groups **348** spaced apart along the X direction, wherein each gate line group **348** comprises two gate lines **346** spaced apart along the X direction, and is located between two adjacent ones of the semiconductor pillars **316** along the X direction, and the first cavity **334** is located between the two gate lines **346** in the same gate line group **348**.

[0182] As shown in FIGS. **7A** and **7C**, in some examples, the semiconductor device **300** further comprises a third cavity **336** located between the two adjacent gate line groups **348** along the X direction and extending along the Y direction, wherein the third cavity **336** is located between the two adjacent ones of the semiconductor pillars **316** along the X direction, and the first cavity **334**, the second cavity **340** and the third cavity **336** are connected.

[0183] As shown in FIG. **7B**, in some examples, each semiconductor pillar **316** extends along a Z direction that is perpendicular to a plane formed by the X direction and the Y direction; the third

cavity **336** has two opposite surfaces along the Z direction, and the surface of the third cavity **336** away from the second cavity **340** is a flat surface.

[0184] As shown in FIG. 7B, in some examples, the semiconductor pillar **316** comprises, along the Z direction, a source **350**, a drain **352** and a channel region **354** between the source **350** and the drain **352**; and the semiconductor device **300** further comprises a gate structure **332** on at least a sidewall of the semiconductor pillar **316**, wherein the gate structures **332** of the plurality of semiconductor pillars **316** arranged along the Y direction are connected to form a gate line **346**.

[0185] Here, the gate structure **332** comprises a gate dielectric layer **342** covering a sidewall of the semiconductor pillar **316** and a gate layer **344** covering the gate dielectric layer **342**.

[0186] As shown in FIG. 7B, in some examples, the bit line **356** is connected with the plurality of drains **352** or sources **350** arranged along the X direction, and the semiconductor device **300** further comprises a memory capacitor connected with the source **350** or the drain **352**. FIG. 7B illustrates that the bit line **356** is connected with the plurality of drains **352** arranged along the X direction.

[0187] With reference to FIGS. 8A to 8C, FIGS. 8A to 8C are cross-sectional structural diagrams of a semiconductor device provided by a second example. FIG. 8A is an XY cross-sectional structural diagram of a semiconductor device provided by a second example, FIG. 8B is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. 8A, and FIG. 8C is an XZ cross-sectional structural diagram along a C-C cross-section in FIG. 8A.

[0188] As shown in FIGS. 8A to 8C, in some examples, the semiconductor device **400** further comprises: a third cavity **436** located between two adjacent ones of the gate line groups **448** along the X direction and extending along the Y direction, wherein the third cavity **436** is located between two adjacent ones of the semiconductor pillars **416** along the X direction, and the second cavity **440** and the third cavity **436** are not connected; and a blocking layer **420** at least covering part of the third cavity **436**, wherein at least part of the blocking layer **420** is located between the semiconductor pillar **416** and the third cavity **436**, and at least part of the blocking layer **420** is located between the second cavity **440** and the third cavity **436**.

[0189] As shown in FIG. 8C, in some examples, the semiconductor pillar **416** extends along a Z direction that is perpendicular to a plane formed by the X direction and the Y direction, the third cavity **436** has two opposite surfaces along the Z direction, and the surface of the third cavity **436** away from the second cavity **440** has a convex shape towards a direction away from the second cavity **440**.

[0190] Here, each gate line group **448** comprises two gate lines **446** extending along the Y direction. A first cavity **434** is disposed between the two gate lines **446** in the same gate line group **448**.

[0191] Here, the gate structure **432** comprises a gate dielectric layer **442** covering a sidewall of the semiconductor pillar **416** and a gate layer **444** covering the gate dielectric layer **442**.

[0192] Here, the semiconductor pillar **416** comprises, along the Z direction, a source **450**, a drain **452** and a channel region **454** between the source **450** and the drain **452**. The semiconductor device **400** comprises a bit line **356** connected with the plurality of drains **352** arranged along the X direction, and a memory capacitor connected with the source **350**.

[0193] With reference to FIGS. 9A to 9C, FIGS. 9A to 9C are cross-sectional structural diagrams of a semiconductor device provided by a third example. FIG. 9A is an XY cross-sectional structural diagram of a semiconductor device provided by a third example, FIG. 9B is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. 9A, and FIG. 9C is an XZ cross-sectional structural diagram along a C-C cross-section in FIG. 9A.

[0194] As shown in FIGS. 9A to 9C, examples of the present disclosure provide a semiconductor device. The semiconductor device **500** comprises: a plurality of semiconductor pillars **516** arranged along an X direction and a Y direction, wherein the X direction intersects the Y direction; a plurality of gate line groups **548** spaced apart along the X direction, wherein each of the plurality of gate line groups **548** comprises two gate lines **546** spaced apart along the X direction, and is located

between the two adjacent ones of the plurality of semiconductor pillars **516** along the X direction, and each gate line **546** extends along the Y direction and is connected with the plurality of semiconductor pillars **516** arranged along the Y direction; a third cavity **536** located between two adjacent ones of the plurality of gate line groups **548** along the X direction and extending along the Y direction; a plurality of bit lines **556** spaced apart along the Y direction, wherein each bit line **556** extends along the X direction and is connected with the plurality of semiconductor pillars **516** arranged along the X direction; and a second cavity **540** located between two adjacent ones of the bit lines **556** along the Y direction and extending along the X direction.

[0195] As shown in FIGS. **9A** and **9C**, in some examples, the third cavity **536** is located between the two adjacent ones of the semiconductor pillars **516** along the X direction, and the second cavity **540** and the third cavity **536** are connected.

[0196] As shown in FIG. **9B**, in some examples, each semiconductor pillar **516** extends along a Z direction that is perpendicular to a plane formed by the X direction and the Y direction; the third cavity **536** has two opposite surfaces along the Z direction, and the surface of the third cavity **536** away from the second cavity **540** is a flat surface.

[0197] As shown in FIGS. **9A** and **9B**, in some examples, the semiconductor pillar **516** comprises, along the Z direction, a source **550**, a drain **552** and a channel region **554** between the source **550** and the drain **552**, and the semiconductor device **500** further comprises a gate structure **532** on at least a sidewall of the semiconductor pillar **516**, wherein the gate structures **532** of the plurality of semiconductor pillars **516** arranged along the Y direction are connected to form a gate line **546**.

[0198] Here, the gate structure **532** comprises a gate dielectric layer **542** covering the sidewall of the semiconductor pillar **516** and a gate layer **544** covering the gate dielectric layer **542**.

[0199] As shown in FIG. **9B**, in some examples, the bit line **556** is connected with the plurality of drains **552** or sources **550** arranged along the X direction, and the semiconductor device **500** further comprises a memory capacitor connected with the source **550** or the drain **552**. FIG. **9B** illustrates that the bit line **556** is connected with the plurality of drains **552** arranged along the X direction.

[0200] With reference to FIGS. **10A** to **10C**, FIGS. **10A** to **10C** are cross-sectional structural diagrams of a semiconductor device provided by a fourth example. FIG. **10A** is an XY cross-sectional structural diagram of a semiconductor device provided by a fourth example, FIG. **10B** is an XZ cross-sectional structural diagram along a B-B cross-section in FIG. **10A**, and FIG. **10C** is an XZ cross-sectional structural diagram along a C-C cross-section in FIG. **10A**.

[0201] As shown in FIGS. **10A** to **10C**, in some examples, the third cavity **636** is located between two adjacent ones of the semiconductor pillars **616** along the X direction, and the second cavity **640** and the third cavity **636** are not connected; and the semiconductor device **600** further comprises a blocking layer **620** at least covering part of the third cavity **636**, wherein at least part of the blocking layer **620** is located between the semiconductor pillar **616** and the third cavity **636**, and at least part of the blocking layer **620** is located between the second cavity **640** and the third cavity **636**.

[0202] As shown in FIG. **10C**, in some examples, each semiconductor pillar **616** extends along a Z direction that is perpendicular to a plane formed by the X direction and the Y direction, the third cavity **636** has two opposite surfaces along the Z direction, and the surface of the third cavity **636** away from the second cavity **640** has a convex shape towards a direction away from the second cavity **640**.

[0203] Here, each gate line group **648** comprises two gate lines **646** extending along the Y direction.

[0204] Here, the gate structure **632** comprises a gate dielectric layer **642** covering the sidewall of the semiconductor pillar **616** and a gate layer **644** covering the gate dielectric layer **642**.

[0205] Here, the semiconductor pillar **616** comprises, along the Z direction, a source **650**, a drain **652** and a channel region **654** between the source **650** and the drain **652**. The semiconductor device **600** comprises a bit line **656** connected with the plurality of drains **652** arranged along the X

direction, and a memory capacitor connected with the source **650**.

[0206] Examples of the present disclosure provide a semiconductor device and a manufacturing method thereof. The semiconductor device comprises: a plurality of semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; a plurality of gate lines spaced apart along the first direction, wherein each of the plurality of gate lines extends along the second direction and is connected with the plurality of semiconductor pillars arranged along the second direction; a first cavity located between two adjacent ones of the plurality of gate lines along the first direction and extending along the second direction; a plurality of bit lines spaced apart along the second direction, wherein each of the plurality of bit lines extends along the first direction and is connected with the plurality of semiconductor pillars arranged along the first direction; and a second cavity located between two adjacent ones of the plurality of bit lines along the second direction and extending along the first direction, wherein the second cavity and the first cavity are connected. In the examples of the present disclosure, the first cavity between two adjacent ones of the gate lines and the second cavity between two adjacent ones of the bit lines are connected, which is beneficial to reduce the capacitive coupling effect between the adjacent gate lines and between the adjacent bit lines, so as to improve the performance of the semiconductor device.

[0207] In view of this, examples of the present disclosure provide a semiconductor device and a manufacturing method thereof.

[0208] In order to achieve the above purpose, the technical solution of the present disclosure is as follows:

[0209] In a first aspect, examples of the present disclosure provide a semiconductor device which comprises a plurality of semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; a plurality of gate lines spaced apart along the first direction, wherein each of the plurality of gate lines extends along the second direction and is connected with the plurality of semiconductor pillars arranged along the second direction; a first cavity located between two adjacent ones of the plurality of gate lines along the first direction and extending along the second direction; a plurality of bit lines spaced apart along the second direction, wherein each of the plurality of bit lines extends along the first direction and is connected with the plurality of semiconductor pillars arranged along the first direction; and a second cavity located between two adjacent ones of the plurality of bit lines along the second direction and extending along the first direction, wherein the second cavity and the first cavity are connected.

[0210] In some examples, the plurality of gate lines comprise a plurality of gate line groups spaced apart along the first direction, and each of the plurality of gate line groups comprises two gate lines spaced apart along the first direction, and is located between two adjacent ones of the plurality of semiconductor pillars along the first direction. The first cavity is located between the two gate lines in the same gate line group.

[0211] In some examples, the semiconductor device further comprises a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction, wherein the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the first cavity, the second cavity and the third cavity are connected.

[0212] In some examples, each of the semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity is a flat surface.

[0213] In some examples, the semiconductor device further comprises a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction, wherein the third cavity is located between two adjacent ones

of the plurality of semiconductor pillars along the first direction, and the second cavity and the third cavity are not connected; and a blocking layer at least covering part of the third cavity, wherein at least part of the blocking layer is located between the semiconductor pillar and the third cavity, and at least part of the blocking layer is located between the second cavity and the third cavity.

[0214] In some examples, each of the semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity has a convex shape towards a direction away from the second cavity.

[0215] In some examples, each of the semiconductor pillars comprises, along the third direction, a source, a drain and a channel region between the source and the drain. The semiconductor device further comprises a gate structure located on at least a sidewall of the semiconductor pillar, wherein the gate structures of the plurality of semiconductor pillars arranged along the second direction are connected to form the gate line.

[0216] In some examples, the bit line is connected with a plurality of the drains or the sources arranged along the first direction, and the semiconductor device further comprises a memory capacitor connected with the source or the drain.

[0217] In a second aspect, examples of the present disclosure provide a manufacturing method of a semiconductor device, which comprises: providing a semiconductor structure comprising a semiconductor layer as well as a plurality of first trenches extending along a first direction and a plurality of second trenches extending along a second direction in the semiconductor layer, wherein the first trenches are filled with a first sacrificial layer, and the first direction intersects the second direction; forming an initial gate structure and a second sacrificial layer filling the second trench in the second trench; removing part of the semiconductor layer to expose the first sacrificial layer; etching the first sacrificial layer to form a plurality of third trenches extending along the first direction and arranged along the second direction; removing part of the initial gate structure and the second sacrificial layer in the second trench through the third trench to divide one initial gate structure into two gate structures and to form a first cavity; and forming a first covering layer sealing the third trench to form a second cavity in the third trench, wherein the first cavity and the second cavity are connected.

[0218] In some examples, the forming the initial gate structure and the second sacrificial layer filling the second trench in the second trench comprises: forming a first dielectric layer covering a sidewall and bottom of the second trench; forming the initial gate structure covering the sidewall and the bottom of the second trench; filling a sacrificial material in the second trench to form the second sacrificial layer; back-etching part of the initial gate structure and the second sacrificial layer, wherein surfaces of the initial gate structure and the second sacrificial layer are lower than a surface of the semiconductor layer; and filling a dielectric material in the second trench.

[0219] In some examples, the semiconductor structure further comprises a plurality of fourth trenches extending along the second direction, and the second trenches and the fourth trenches are arranged alternately along the first direction, wherein depths of the first trenches are greater than depths of the fourth trenches.

[0220] In some examples, the method further comprises: forming a second dielectric layer covering a sidewall and bottom of the fourth trench; filling a sacrificial material in the fourth trench to form a third sacrificial layer; back-etching part of the third sacrificial layer, wherein a surface of the third sacrificial layer is lower than a surface of the semiconductor layer; and filling a dielectric material in the fourth trench.

[0221] In some examples, at the same time as removing part of the initial gate structure and the second sacrificial layer in the second trench through the third trench to divide one initial gate structure into two gate structures and to form the first cavity, the method further comprises: removing the third sacrificial layer in the fourth trench through the third trench to form a third

cavity in the fourth trench, wherein the first cavity, the second cavity and the third cavity are connected.

[0222] In some examples, the method further comprises: forming a second dielectric layer covering the sidewall and the bottom of the fourth trench; forming a blocking layer covering the sidewall and the bottom of the fourth trench, back-etching part of the blocking layer, wherein a surface of the blocking layer is lower than a surface of the semiconductor layer; and forming a second covering layer sealing the fourth trench to form a third cavity in the fourth trench, wherein the second cavity and the third cavity are not connected.

[0223] In a third aspect, examples of the present disclosure provide a semiconductor device comprising a plurality of semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; a plurality of gate line groups spaced apart along the first direction, wherein each of the plurality of gate line groups comprises two gate lines spaced apart along the first direction, and is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and each of the gate lines extends along the second direction and is connected with the plurality of semiconductor pillars arranged along the second direction; a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction; a plurality of bit lines spaced apart along the second direction, wherein each of the plurality of bit lines extends along the first direction and is connected with the plurality of semiconductor pillars arranged along the first direction; and a second cavity located between two adjacent ones of the plurality of bit lines along the second direction and extending along the first direction.

[0224] In some examples, the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the second cavity and the third cavity are connected.

[0225] In some examples, each of the plurality of semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity is a flat surface.

[0226] In some examples, the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the second cavity and the third cavity are not connected. The semiconductor device further comprises a blocking layer at least covering part of the third cavity, wherein at least part of the blocking layer is located between the semiconductor pillar and the third cavity, and at least part of the blocking layer is located between the second cavity and the third cavity.

[0227] In some examples, each of the plurality of semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity has a convex shape towards a direction away from the second cavity.

[0228] In some examples, each of the semiconductor pillars comprises, along the third direction, a source, a drain and a channel region between the source and the drain. The semiconductor device further comprises a gate structure located on at least a sidewall of the semiconductor pillar, wherein the gate structures of the plurality of semiconductor pillars arranged along the second direction are connected to form the gate line.

[0229] In some examples, the bit line is connected with a plurality of the drains or the sources arranged along the first direction, and the semiconductor device further comprises a memory capacitor connected with the source or the drain.

[0230] Examples of the present disclosure provide a semiconductor device and a manufacturing method thereof. In the examples of the present disclosure, the first cavity between two adjacent ones of the gate lines and the second cavity between two adjacent ones of the bit lines are connected, which is beneficial to reduce a capacitive coupling effect between the adjacent gate

lines and between the adjacent bit lines, so as to improve the performance of the semiconductor device.

[0231] It is to be understood that, references to “one example” or “an example” throughout the specification indicate that particular features, structures, or characteristics related to the example are comprised in at least one example of the present disclosure. Therefore, “in one example” or “in an example” presented throughout this specification does not necessarily refer to the same example. Furthermore, these particular features, structures, or characteristics may be incorporated in one or more examples in any suitable manner. It is to be understood that, in various examples of the present disclosure, sequence numbers of the above processes do not indicate a sequence of execution, and the sequence of execution of various processes shall be determined by functionalities and intrinsic logics thereof, and shall constitute no limitation on the implementation of examples of the present disclosure. The above sequence numbers of the examples of the present disclosure are only for description, and do not represent advantages and disadvantages of the examples.

[0232] The above descriptions are merely some implementations of the present disclosure, and are not intended to limit the protection scope of the present disclosure. Any equivalent structure transformation made by using the contents of the specification and the drawings of the present disclosure under the inventive concept of the present disclosure, or direct/indirect application to other related technical fields are encompassed within the protection scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a plurality of semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; a plurality of gate lines spaced apart along the first direction, wherein each of the plurality of gate lines extends along the second direction and is connected with the plurality of semiconductor pillars arranged along the second direction; a first cavity located between two adjacent ones of the plurality of gate lines along the first direction and extending along the second direction; a plurality of bit lines spaced apart along the second direction, wherein each of the plurality of bit lines extends along the first direction and is connected with the plurality of semiconductor pillars arranged along the first direction; and a second cavity located between two adjacent ones of the plurality of bit lines along the second direction and extending along the first direction, wherein the second cavity and the first cavity are connected.
2. The semiconductor device of claim 1, wherein the plurality of gate lines include a plurality of gate line groups spaced apart along the first direction, and each of the plurality of gate line groups includes two gate lines spaced apart along the first direction, and is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the first cavity is located between the two gate lines in a same gate line group.
3. The semiconductor device of claim 2, further including a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction, wherein the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the first cavity, the second cavity and the third cavity are connected.
4. The semiconductor device of claim 3, wherein each of the semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, and the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity is a flat surface.
5. The semiconductor device of claim 2, further including: a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction, wherein the third cavity is located between two adjacent ones of the plurality of

semiconductor pillars along the first direction, and the second cavity and the third cavity are not connected; and a blocking layer at least covering part of the third cavity, wherein at least part of the blocking layer is located between a semiconductor pillar and the third cavity, and at least part of the blocking layer is located between the second cavity and the third cavity.

6. The semiconductor device of claim 5, wherein each of the plurality of semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, and the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity has a convex shape towards a direction away from the second cavity.

7. The semiconductor device of claim 4, wherein each of the semiconductor pillars includes, along the third direction, a source, a drain and a channel region between the source and the drain, and the semiconductor device further includes a gate structure located on at least a sidewall of a semiconductor pillar, wherein the gate structures of the plurality of semiconductor pillars arranged along the second direction are connected to form a gate line.

8. The semiconductor device of claim 7, wherein a bit line is connected with a plurality of drains or sources arranged along the first direction, and the semiconductor device further includes a memory capacitor connected with the source or the drain.

9. A manufacturing method of a semiconductor device, comprising: providing a semiconductor structure including a semiconductor layer as well as a plurality of first trenches extending along a first direction and a plurality of second trenches extending along a second direction in the semiconductor layer, wherein the first trenches are filled with a first sacrificial layer, and the first direction intersects the second direction; forming an initial gate structure and a second sacrificial layer filling the second trench in the second trench; removing part of the semiconductor layer to expose the first sacrificial layer; etching the first sacrificial layer to form a plurality of third trenches extending along the first direction and arranged along the second direction; removing part of the initial gate structure and the second sacrificial layer in the second trench through the third trench to divide one initial gate structure into two gate structures and to form a first cavity; and forming a first covering layer sealing the third trench to form a second cavity in the third trench, wherein the first cavity and the second cavity are connected.

10. The manufacturing method of claim 9, wherein the forming the initial gate structure and the second sacrificial layer filling the second trenches in the second trenches further includes: forming a first dielectric layer covering a sidewall and bottom of the second trenches; forming the initial gate structure covering the sidewall and the bottom of the second trenches; filling a sacrificial material in the second trenches to form the second sacrificial layer; back-etching part of the initial gate structure and the second sacrificial layer, wherein surfaces of the initial gate structure and the second sacrificial layer are lower than a surface of the semiconductor layer; and filling a dielectric material in the second trenches.

11. The manufacturing method of claim 9, wherein the semiconductor structure further includes a plurality of fourth trenches extending along the second direction, and the second trenches and the fourth trenches are arranged alternately along the first direction, wherein depths of the first trenches are greater than depths of the fourth trenches.

12. The manufacturing method of claim 11, further including: forming a second dielectric layer covering a sidewall and bottom of the fourth trenches; filling a sacrificial material in the fourth trenches to form a third sacrificial layer; back-etching part of the third sacrificial layer, wherein a surface of the third sacrificial layer is lower than a surface of the semiconductor layer; and filling a dielectric material in the fourth trenches.

13. The manufacturing method of claim 12, wherein, at the same time as removing part of the initial gate structure and the second sacrificial layer in the second trench through the third trenches to divide one initial gate structure into two gate structures and to form the first cavity, the method further includes removing the third sacrificial layer in the fourth trench through the third trenches

to form a third cavity in the fourth trench, wherein the first cavity, the second cavity and the third cavity are connected.

14. The manufacturing method of claim 11, further including: forming a second dielectric layer covering a sidewall and bottom of the fourth trenches; forming a blocking layer covering the sidewall and the bottom of the fourth trenches; back-etching part of the blocking layer, wherein a surface of the blocking layer is lower than a surface of the semiconductor layer; and forming a second covering layer sealing the fourth trenches to form a third cavity in the fourth trenches, wherein the second cavity and the third cavity are not connected.

15. A semiconductor device, comprising: a plurality of semiconductor pillars arranged along a first direction and a second direction, wherein the first direction intersects the second direction; a plurality of gate line groups spaced apart along the first direction, wherein each of the plurality of gate line groups includes two gate lines spaced apart along the first direction, and is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and each of the gate lines extends along the second direction and is connected with the plurality of semiconductor pillars arranged along the second direction; a third cavity located between two adjacent ones of the plurality of gate line groups along the first direction and extending along the second direction; a plurality of bit lines spaced apart along the second direction, wherein each of the plurality of bit lines extends along the first direction and is connected with the plurality of semiconductor pillars arranged along the first direction; and a second cavity located between two adjacent ones of the plurality of bit lines along the second direction and extending along the first direction.

16. The semiconductor device of claim 15, wherein the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the second cavity and the third cavity are connected.

17. The semiconductor device of claim 16, wherein each of the plurality of semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, and the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity is a flat surface.

18. The semiconductor device of claim 15, wherein the third cavity is located between two adjacent ones of the plurality of semiconductor pillars along the first direction, and the second cavity and the third cavity are not connected, and the semiconductor device further includes a blocking layer at least covering part of the third cavity, wherein at least part of the blocking layer is located between a semiconductor pillar and the third cavity, and at least part of the blocking layer is located between the second cavity and the third cavity.

19. The semiconductor device of claim 18, wherein each of the plurality of semiconductor pillars extends along a third direction that is perpendicular to a plane formed by the first direction and the second direction, and the third cavity has two opposite surfaces along the third direction, and a surface of the third cavity away from the second cavity has a convex shape towards a direction away from the second cavity.

20. The semiconductor device of claim 17, wherein each of the semiconductor pillars includes, along the third direction, a source, a drain and a channel region between the source and the drain, and the semiconductor device further includes a gate structure located on at least a sidewall of a semiconductor pillar, wherein the gate structures of the plurality of semiconductor pillars arranged along the second direction are connected to form a gate line, and wherein a bit line is connected with a plurality of drains or sources arranged along the first direction, and the semiconductor device further includes a memory capacitor connected with the source or the drain.
